

FCC3 232 CHAP 3 COMBINED STRESS

3.1 INTRODUCTION

Combined stress occurs in structural members which are subjected to **combined loadings**. Structural members which are subjected to combined loadings include: -

1. **Eccentrically loaded columns**: These are columns where the compressive load acts at an eccentricity with respect to the column longitudinal axis.
2. **Eccentrically loaded foundations**: These are foundation slabs where the column load acts at an eccentricity with respect to the slab centroid axes.
3. **Shaft-beams**: These are shafts which are subjected to both twisting and bending.

Columns and **foundations** are structural members which are normally designed to carry the applied loads by **axial compression**. An **eccentric load** will induce both compression and bending. The **principle of superposition** is used to express the **resultant stress** due to the combined **axial compression** and **bending**. If the eccentricity is large, the **resultant stress** at some part of the section may become **tensile**. The member is said to suffer **reverse stress**. In order to avoid reverse stresses, the applied eccentric load must act within an area of the section called the **section kern**.

Shafts are structural members which are predominantly subjected to torque. A shaft-beam is a shaft which is also subjected to bending. The **maximum normal stress** and **maximum shear stress** due to combined **bending** and **twisting** are expressed in terms of the **equivalent bending** and **equivalent twisting moments**.

The following **assumptions** shall be made.

1. The structural material is **homogeneous, isotropic, linearly elastic**, and is not stressed beyond the **elastic limit**.
2. The member section is **prismatic** and the **stresses are uniform** across the **width of the section** at any level from the neutral axis
3. The principle of **superposition** is **valid** so that the **resultant stress** at any point can be obtained as the **algebraic sum** of the **individual stress types** at that point.

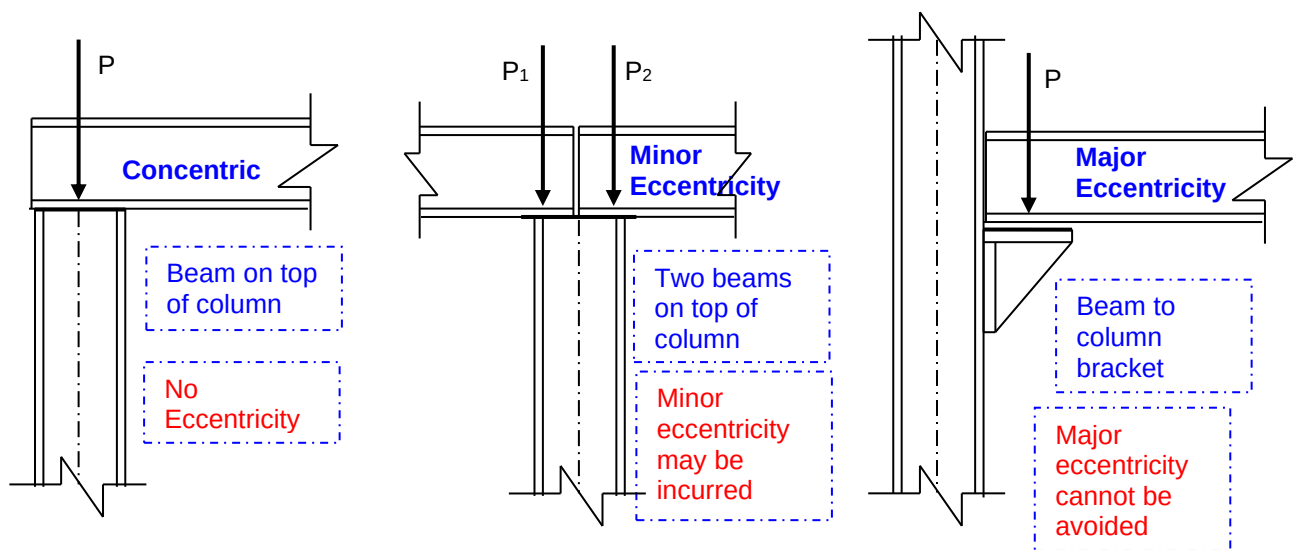
SIGN CONVENTION: **compressive stress** will be designated **positive**
tensile stress will be designated **negative**.

3.2 ECCENTRICALLY LOADED COLUMNS

3.2.1 Causes of Eccentric Loads

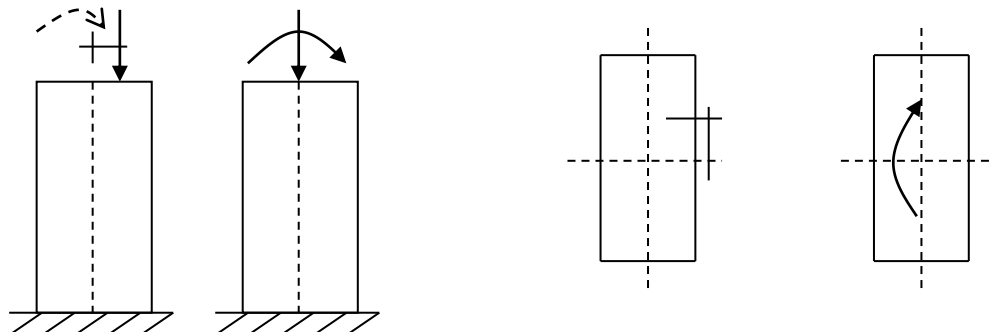
In reinforced concrete construction, the column and beam joints are constructed **monolithically**. The load from the beams is therefore transferred to the column concentrically or with minor eccentricity.

However, in steel construction, the column and beam joints are constructed in **modular** manner by using beam-to-column brackets. The load from the beam may be transferred to the column **concentrically**, with **minor eccentricity** or with **major eccentricity**. The diagram below illustrates the three levels of eccentricity in steel beam-to-column joints.



3.2.2 Resultant stress

Consider the column shown below where the applied axial load P acts at an eccentricity e_x with respect to the section x-axis. The load scenario is the same as the load P acting axially plus a bending moment equal to Pe_x acting about the x-axis.

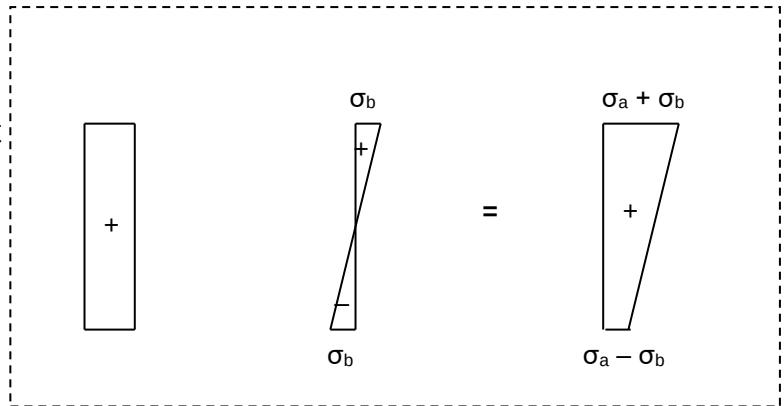


The **resultant stress** at any point on the cross section is the **algebraic sum** of the **axial stress** and the **flexural stress**.

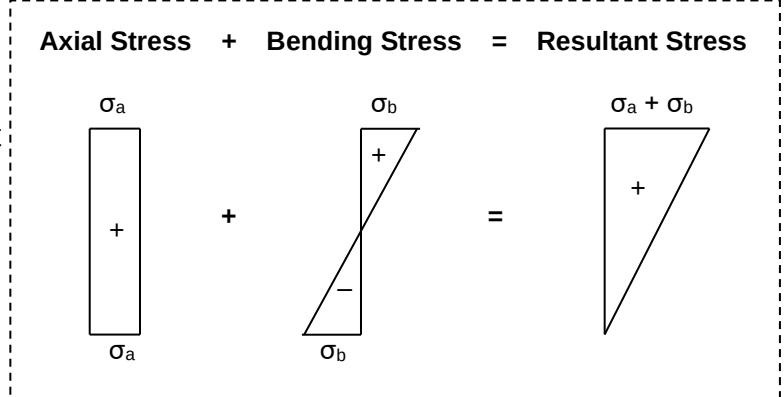
3.2.3 Resultant Stress Distribution

The resultant stress distribution over the cross section can be plotted by superposition. Depending on the amounts of axial and flexural stress, three scenarios are possible.

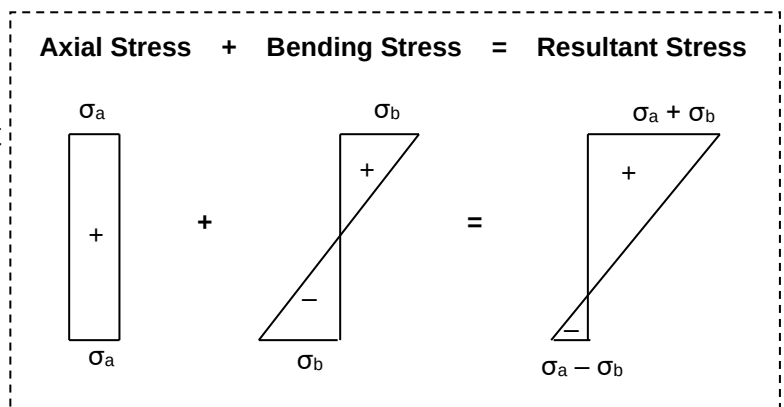
Scenario 1: If the bending stress σ_b at the extreme fibers is **less than** the axial stress σ_a , then the resultant stress σ remains **positive** over the entire cross section.



Scenario 2: If the bending stress σ_b at the extreme fibers is **equal to** the axial stress σ_a , then the resultant stress σ at the bottom fibers becomes **zero**.



Scenario 3: If the bending stress σ_b at the extreme fibers is **more than** the axial stress σ_a , then the resultant stress at the bottom fibers becomes **negative**. **Reverse stress is defined as a condition where the resultant stress at some point in the section becomes negative.**



The resultant stress at any point on the cross section is given by **algebraic summation**

$$\sigma = \sigma_a \pm \sigma_b = P/A \pm P e_x y/l_x \quad \text{-----} \quad \text{Eqn 1}$$

In general, if the load is eccentric with respect to both the x-axis and the y-axis, then the resultant stress at any point on the cross section is given by

$$\sigma = P/A \pm P e_x y/l_x \pm P e_y x/l_y \quad \text{-----} \quad \text{Eqn 1*}$$

3.3 ECCENTRICALLY LOADED FOUNDATIONS

3.3.1 Bearing Pressure

Foundation slabs (commonly called **column footings**) are structural members which are designed to spread the load from columns to the earth material below the building.

The earth material below the footing resists the compression load by developing **bearing pressure** in order to equilibrate the applied load. The bearing pressure beneath the footing is usually calculated in **kN/m²** and is the same as the stress induced on the footing if the footing was considered as a member cross section.

Ideally, the column load should be placed at the centroid of the footing section so that the bearing pressure induced beneath the footing is uniform over the whole section. However, due to design constraints, the column load may be placed at an eccentricity with respect to either the x-axis, or the y-axis or both x- and y-axes. Eqn 1 is used to evaluate the resultant bearing pressure at each corner of the footing. The **bearing pressure distribution** is then plotted **for all sides of the footing**.

Column footings are usually massive structural members, and the **self weight** of the footing may contribute significantly to the **axial load** and therefore the **bearing pressure** induced beneath it. However, in this chapter, the **self weight shall be ignored**.

Consider the column footing shown below and suppose it is required to plot the bearing pressure on all four sides.

The bearing pressure is calculated at each corner using Eqn 1*

$$\text{At A: } \sigma_A = P/A - P e_x (d/2)/I_x + P e_y (b/2)/I_y$$

$$\text{At B: } \sigma_B = P/A + P e_x (d/2)/I_x + P e_y (b/2)/I_y$$

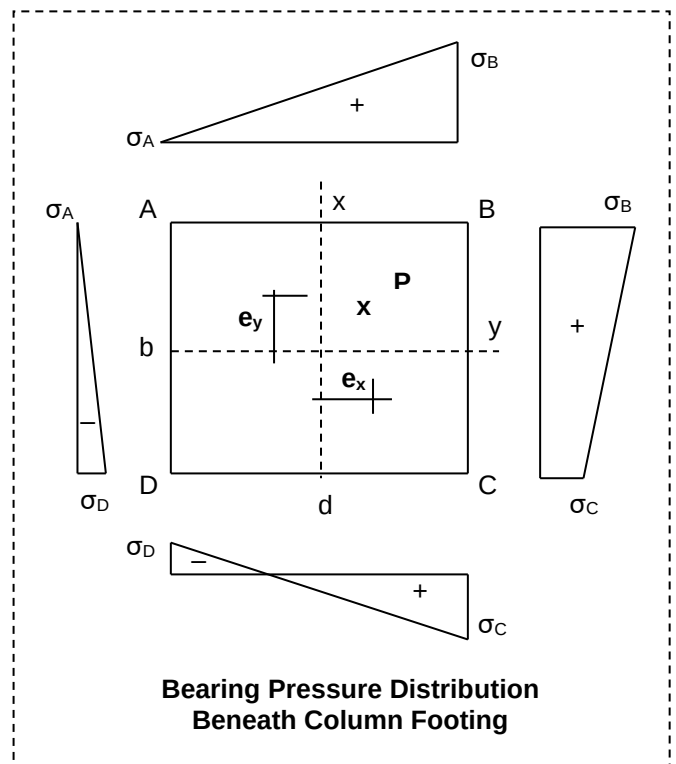
$$\text{At C: } \sigma_C = P/A + P e_x (d/2)/I_x - P e_y (b/2)/I_y$$

$$\text{At D: } \sigma_D = P/A - P e_x (d/2)/I_x - P e_y (b/2)/I_y$$

Assuming $\sigma_A = \text{zero}$, $\sigma_B = +ve$

$$\sigma_C = +ve \text{ and } \sigma_D = -ve$$

The bearing pressure distributions are plotted assuming linear distribution between the corner points, as shown.



3.3.2 The Middle Third Kern Rule

Consider the **eccentrically loaded rectangular section** shown below. If there is to be no reverse stress along the bottom fibers, then using Eqn 1:

$$\sigma = P/A - P e_x (d/2)/I_x \geq 0 \quad \text{OR} \quad e_x \leq 2I_x/Ad$$

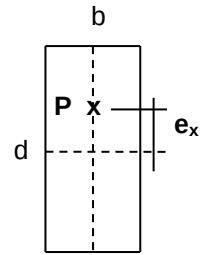
Noting that $I_x = bd^3/12$ and $A = bd$, then

$$e_x \leq d/6 \quad \text{-----} \quad \text{Eqn 2}$$

Eqn 2 gives the limiting eccentricity if there is to be no reverse stress.

Similarly, it can be shown that when the load is eccentric with respect

to the y-axis, the limiting eccentricity is given by $e_y \leq b/6$ ----- Eqn 2*

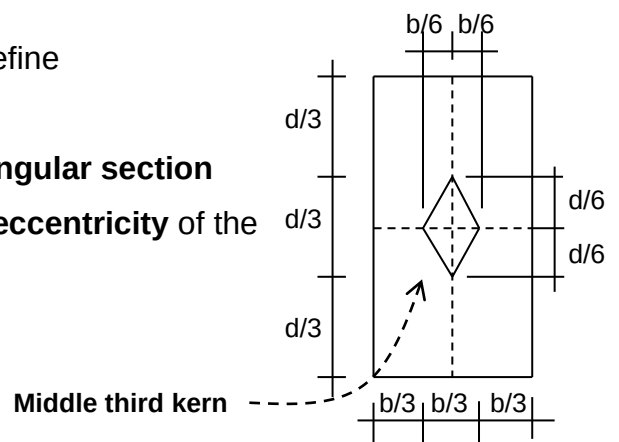


Eqns 2 and 2* can be plotted on the section to define the **middle third kern** of the section.

The **middle third kern rule** states that if a **rectangular section** is not to experience **reverse stresses**, then the **eccentricity** of the applied load must satisfy Eqns 2 and 2*

$$e_x \leq \pm d/6 \quad \text{-----} \quad \text{Eqn 2}$$

And $e_y \leq \pm b/6$ ----- Eqn 2*



3.3.3 The Middle Quarter Kern Rule

Consider the **eccentrically loaded circular section** shown below. If there is to be no reverse stress along the bottom fibers, then using Eqn 1:

$$\sigma = P/A - P e_x (D/2)/I_x \geq 0 \quad \text{OR} \quad e_x \leq 2I_x/AD$$

Noting that $I_x = \pi D^4/64$ and $A = \pi D^2/4$, then

$$e_x \leq D/8 \quad \text{-----} \quad \text{Eqn 3}$$

Eqn 3 gives the limiting eccentricity if there is to be no reverse stress.

Similarly, it can be shown that when the load is eccentric with respect

to the y-axis, then $e_y \leq D/8$ ----- Eqn 3*

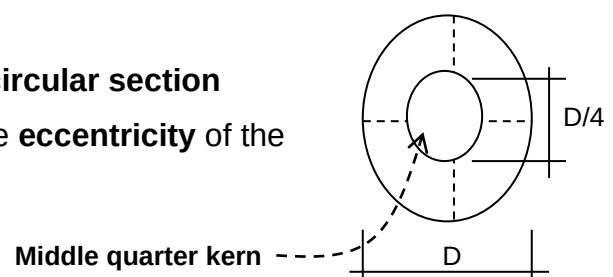
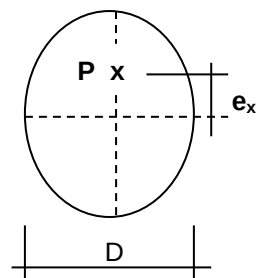
Eqns 3 and 3* can be plotted on the section to define

the **middle quarter kern** of the section.

The **middle quarter kern rule** states that if a **circular section** is not to experience **reverse stresses**, then the **eccentricity** of the applied load must satisfy Eqns 3 and 3*

$$e_x \leq \pm D/8 \quad \text{-----} \quad \text{Eqn 3}$$

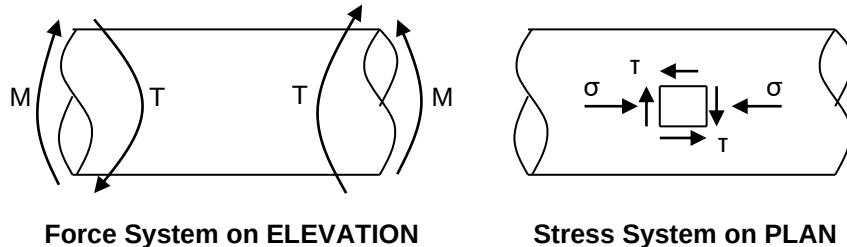
And $e_y \leq \pm D/8$ ----- Eqn 3*



3.4 SHAFT-BEAMS

3.4.1 Equivalent Bending Moment

Consider a solid circular shaft subjected to a bending moment **M** and a twisting moment **T**, which act simultaneously. The stresses on an element on the extreme top fibers of the section are shown alongside.



The normal and shear stresses are given by: $\sigma = My/I$ and $\tau = Tr/J$

Noting that $I = \pi D^4/64$ $J = \pi D^4/32$ $y = D/2$ and $r = D/2$

The section modulus $Z = I/y = \pi D^3/32$ and $J/r = \pi D^3/16 = 2Z$

The normal and shear stresses are given by: $\sigma = M/Z$ and $\tau = T/2Z$

The maximum principal normal stress is given by $\sigma_{\max} = \frac{1}{2}[\sigma + \sqrt{(\sigma^2 + 4\tau^2)}]$

Substituting for σ and τ in terms of M , T and Z , $\sigma_{\max} = \frac{1}{2}[M/Z + \sqrt{(M/Z)^2 + (T/Z)^2}]$

Rearrange to express $\sigma_{\max} Z = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$

The **equivalent bending moment** is defined as $M_E = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$ ----- Eqn 4

The maximum normal stress is therefore given by $\sigma_{\max} = M_E/Z$ ----- Eqn 5

The **equivalent bending moment** is defined as that bending moment which, if acting alone, would induce the same **maximum normal stress** as a bending moment and twisting moment combined.

3.4.2 Equivalent Twisting Moment

Consider the solid circular shaft subjected to a bending moment **M** and a twisting moment **T**, as discussed in **section 3.4.1** above.

The maximum principal shear stress is given by $\tau_{\max} = \frac{1}{2}\sqrt{(\sigma^2 + 4\tau^2)}$

Substituting for σ and τ in terms of M , T and Z , $\tau_{\max} = \frac{1}{2}\sqrt{[(M/Z)^2 + (T/Z)^2]}$

Rearrange to express $\tau_{\max} (2Z) = \sqrt{(M^2 + T^2)}$

The **equivalent twisting moment** is defined as $T_E = \sqrt{(M^2 + T^2)}$ ----- Eqn 6

The maximum shear stress is therefore given by $\tau_{\max} = T_E/2Z$ ----- Eqn 7

The **equivalent twisting moment** is defined as that twisting moment which, if acting alone, would induce the same **maximum shear stress** as a bending moment and twisting moment combined.

3.5 WORKED EXAMPLES

Example 1: The rectangular timber column section shown below is subjected to a compressive load $P = 96 \text{ kN}$ acting at an eccentricity $e_x = 50 \text{ mm}$ with respect to the x-axis. Plot the resultant stress distribution over the section indicating the peak values.

Solution [$I_x = bd^3/12 = 115.2 \times 10^6 \text{ mm}^4$]

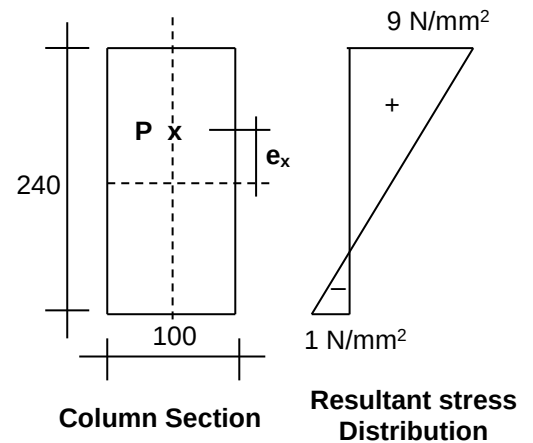
Using Eqn 1 the resultant stress is given by

At the top: $\sigma_{\text{top}} = P/A + P e_x (d/2)/I_x = 9 \text{ N/mm}^2$

At the bottom: $\sigma_{\text{bot}} = P/A - P e_x (d/2)/I_x = -1 \text{ N/mm}^2$

The resultant stress distribution is shown.

[The resultant stress is assumed to be uniform across the section width, since there is no eccentricity with respect to the y-axis]



Example 2: A universal column section is subjected to an axial load $P_1 = 360 \text{ kN}$ and two eccentric loads $P_2 = 246 \text{ kN}$ and $P_3 = 127 \text{ kN}$ as shown below. If $e_y = 60 \text{ mm}$ determine the eccentricity e_x that will make the resultant stress at D be zero. Plot the resultant stress distribution on all sides of the section indicating the peak values.

Solution [$\Sigma P = P_1 + P_2 + P_3 = 733 \text{ kN}$

$$A = 13.6 \times 10^3 \text{ mm}^2$$

$$I_x = 129.8 \times 10^6 \text{ mm}^4$$

$$I_y = 46.2 \times 10^6 \text{ mm}^4]$$

Using Eqn 1*

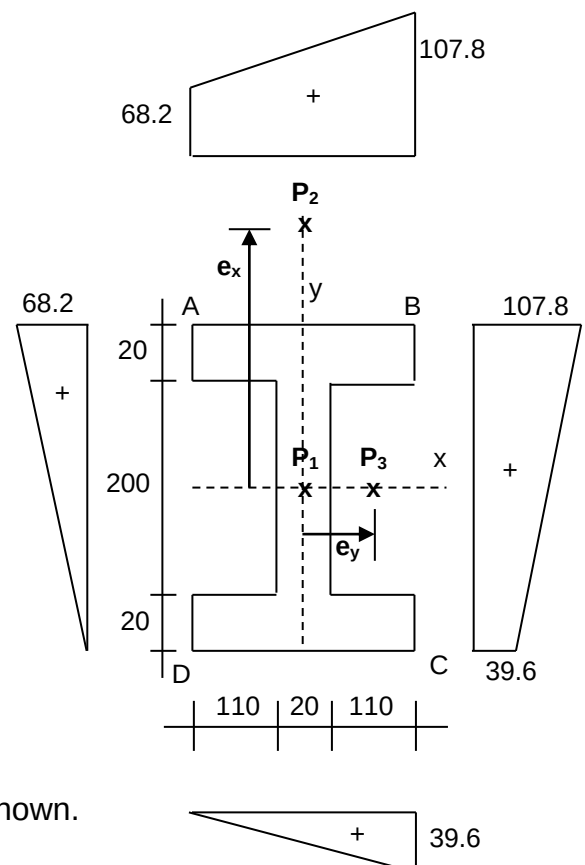
At D: $\sigma_D = \Sigma P/A - P_2 e_x (120)/I_x - P_3 e_y (120)/I_y = 0$

Solving $e_x = 150 \text{ mm}$

At A: $\sigma_A = \Sigma P/A + P_2 e_x (120)/I_x - P_3 e_y (120)/I_y$
 $= 68.2 \text{ N/mm}^2$

At B: $\sigma_B = \Sigma P/A + P_2 e_x (120)/I_x + P_3 e_y (120)/I_y$
 $= 107.8 \text{ N/mm}^2$

At C: $\sigma_C = \Sigma P/A - P_2 e_x (120)/I_x + P_3 e_y (120)/I_y$
 $= 39.6 \text{ N/mm}^2$

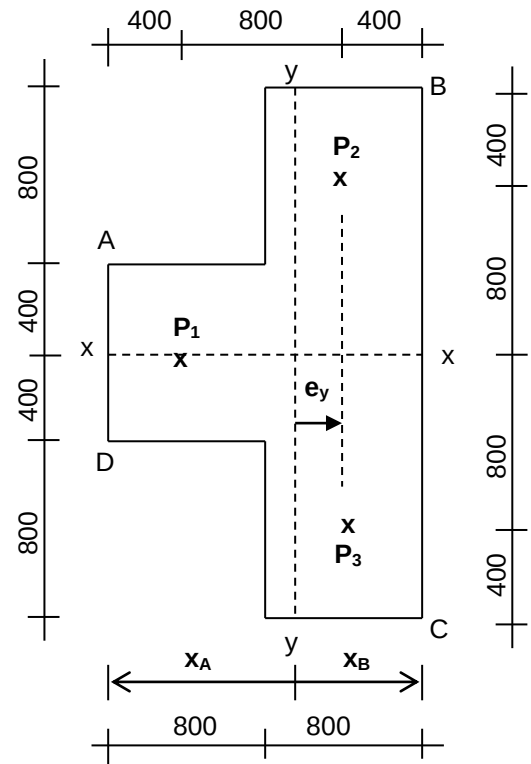


The resultant stress distribution (in N/mm²) is shown.

Example 3: A foundation slab is subjected to three column loads $P_1 = 232 \text{ kN}$, $P_2 = 384 \text{ kN}$ and $P_3 = 384 \text{ kN}$ arranged as shown below. Determine the eccentricity e_y of P_2 and P_3 with respect to the y-axis. Evaluate the bearing pressure beneath the points A, B, C and D.

Solution

$$\begin{aligned}\Sigma P &= 1000 \text{ kN} & A &= 2.56 \text{ m}^2 \\ x_B &= 0.6 \text{ m} & e_y &= 0.2 \text{ m} \\ I_x &= 955.73 \times 10^{-3} \text{ m}^4 & I_y &= 443.73 \times 10^{-3} \text{ m}^4 \\ \sigma_A &= \Sigma P/A + P_1 (0.6)(1.0)/I_y \\ &\quad + P_2 (0.8)(0.4)/I_x - P_2 (0.2)(1.0)/I_y \\ &\quad - P_3 (0.8)(0.4)/I_x - P_3 (0.2)(1.0)/I_y = 358 \text{ kN/m}^2 \\ \sigma_B &= \Sigma P/A - P_1 (0.6)(0.6)/I_y \\ &\quad + P_2 (0.8)(1.2)/I_x + P_2 (0.2)(0.6)/I_y \\ &\quad - P_3 (0.8)(1.2)/I_x + P_3 (0.2)(0.6)/I_y = 410 \text{ kN/m}^2 \\ \sigma_C &= \Sigma P/A - P_1 (0.6)(0.6)/I_y \\ &\quad - P_2 (0.8)(1.2)/I_x + P_2 (0.2)(0.6)/I_y \\ &\quad + P_3 (0.8)(1.2)/I_x + P_3 (0.2)(0.6)/I_y = 410 \text{ kN/m}^2 \\ \sigma_D &= \Sigma P/A + P_1 (0.6)(1.0)/I_y \\ &\quad - P_2 (0.8)(0.4)/I_x - P_2 (0.2)(1.0)/I_y \\ &\quad + P_3 (0.8)(0.4)/I_x - P_3 (0.2)(1.0)/I_y = 358 \text{ kN/m}^2\end{aligned}$$



Example 4: A hollow circular shaft is subjected to a bending moment of 8 kNm and a twisting moment of 4 kNm. If the internal and external diameters of the shaft are 80 mm and 100 mm respectively, evaluate the maximum normal and maximum shear stresses induced in the shaft.

Solution

Given $M = 8 \text{ kNm}$ $T = 4 \text{ kNm}$ $D_o = 100 \text{ mm}$ $D_i = 80 \text{ mm}$

Using Eqns 4 and 6 $M_E = \frac{1}{2}[M + \sqrt{M^2 + T^2}] = 8.5 \text{ kNm}$

$T_E = \sqrt{M^2 + T^2} = 8.9 \text{ kNm}$

And $Z = (\pi/32)(D_o^4 - D_i^4)/D_o = 57.96 \times 10^3 \text{ mm}^3$

Using Eqn 5 $\sigma_{\max} = M_E/Z = 146.7 \text{ N/mm}^2$

Using Eqn 7 $\tau_{\max} = T_E/2Z = 76.8 \text{ N/mm}^2$